Mingyu Li
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Education Background

Tongji University **School of Mechanical Engineering** GPA: 4.23/5 Overall Ranking: 45/138

Major in Mechanical Design, Manufacturing and Automation Sep. 2023 - Present

Core Courses: Mechanical Drawing, Innovative Design of Mechanical Engineering, C/C++ Programming

Research Experience

CAD、Linux、ROS、Python、Embodied Vision

1.Zero-Shot Training Dexterous Hand Generalized Visual Grasping Framework for Open Environments

Core Leader | Sep. 2025 – Present

Project Introduction: This project designed an edge intelligence-based visual servo control framework for dexterous hands applicable to open environments, enabling users to grasp common objects (non-COCO80 categories) without pre-training.

Responsibilities: Undertaken mechanical design and manufacturing of the project, compiled control programs for Linux development boards, constructed and optimized the visual algorithm framework, and delivered project defense reports; Proficient in using software such as Inventor, Mobaxterm and MATLAB; Joined the supervisor's Vision4Robot Lab and read Chinese and English literatures in the relevant field under the guidance of the supervisor.

2. Precise Grasping of Manipulator Under the Constraints of 2D Planar Camera and Embedded System

Team Member | Dec. 2024 - Mar. 2026

Project Introduction: This project optimized the manipulator grasping scheme under the constraints of 2D planar camera and low-computing-power embedded system.

Responsibilities: Conducted mechanical design of the manipulator, and deployed visual algorithms and grasping algorithms on the embedded platform.

3. Simulation and Optimization of Manual Work in Intelligent Manufacturing System Based on ema WD Software Team Member | Apr. 2024 - Jun. 2025

Project Introduction: Established an assembly production line for gear boxes in the ema WD digital twin software, optimized workers' working environment according to ergonomics and conducted quantitative analysis in the software.

Responsibilities: Completed the modeling of gear boxes, learned knowledge related to ergonomics and mastered digital twin related software.

Awards & Achievements

Awards

- The 18th "CAD Cup" - Additive Manufacturing Track (Mechanical Engineering) | **National First Prize**
- 2025 "BQRI Cup" Logistics Technology Innovation Competition | **National Second Prize**
- The 20th National University Student Intelligent Vehicle Competition | **Provincial Third Prize**
- University-level Excellent Student, and 6 other provincial, municipal and university-level awards

Patents & Published Papers

- *Folding Crutch Chair* (Patent No.: 202521174680.9) | Utility Model Patent, First Author
- [Paper Title Omitted] | ICARM Conference, CAA-A, First Author

Research Skills

- **Modeling:** Proficient in Inventor; Skilled in using Solidworks, Fusion360 and Revit.
- **Electrical Engineering:** Master MATLAB design software; Rich practical experience in Simulink and Simscape.
- **Programming:** Master Python; Specialized in developing Linux development boards and deploying visual algorithms; Hands-on experience in ROS development.